

# SAGAR VARMA

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## EDUCATION

### Georgia Institute of Technology

M.S. in Computer Science (OMSCS)

Atlanta, GA

Expected May 2028

### Rutgers University - New Brunswick

B.S. in Computer Science, Minor in Data Science **GPA: 3.5/4.0**

New Brunswick, NJ

Sep. 2021 – May 2025

- Dean's List (Multiple Semesters)

## EXPERIENCE

### Data Engineer (Python, C#, SQL, Azure, OpenAI, Redshift)

May 2025 – Present

Bausch Health

Bridgewater, NJ

- Designed and maintained REST API ingestion pipelines aggregating data spanning 5M+ biomedical articles.
- Engineered market access reconciliation pipelines surfacing discrepancies across rebate and contract datasets.
- Built a scalable RAG ingestion pipeline enabling semantic retrieval and excerpt extraction across verified sources.
- Supported disaster recovery data integration workflows ensuring reporting continuity across Azure environments.
- Provisioned Azure data infrastructure via Bicep IaC with Key Vault, VNet endpoints, and CI/CD pipelines.
- Delivered a unified research platform reducing manual literature review by 40% for R&D teams.

### Data Engineering Intern (Python, SQL, Selenium, ML, OpenAI)

May 2024 – Aug 2024

Bausch & Lomb

Bridgewater, NJ

- Built Python ingestion and transformation pipelines processing 100k+ device log records for downstream analysis.
- Automated Lansweeper data extraction via headless Selenium, refreshing device inventory datasets hourly.
- Developed anomaly detection pipeline using Isolation Forest to flag highest-frequency device error patterns.
- Built clustering pipeline grouping devices by shared failure signatures using k-means for root cause analysis.
- Integrated OpenAI layer to auto-diagnose clustered device failures and surface recommended remediation steps.
- Filtered 100k+ records to critical assets, enabling proactive user outreach ahead of ticket submission.

### Data Engineer Intern (Python, SQL, ETL, Tableau)

May 2023 – Sep 2023

Rutgers University

Piscataway, NJ

- Built Python and SQL ETL pipelines processing multi-year institutional datasets for scheduled reporting systems.
- Automated data validation, transformation, and scheduled refresh jobs across recurring reporting workflows.
- Optimized pipeline logic and query performance, reducing end-to-end processing time by approximately 30%.

### Data Analyst Intern

May 2023 – Aug 2023

Tyger Fit Lab

Caldwell, NJ

- Built SQL reporting workflows tracking transactions, inventory trends, and KPIs for e-commerce operations.
- Developed Power BI dashboards visualizing sales performance and inventory metrics for stakeholder review.

## PROJECTS

### Dark Pool Activity Detection Platform (GitHub)

Apache Kafka, Apache Spark, FastAPI, React, Docker

- Built a real-time market surveillance pipeline ingesting simulated equity trade streams using Apache Kafka.
- Implemented Spark Structured Streaming jobs for windowed aggregations and trade volume anomaly detection.
- Delivered real-time alerts to a React dashboard via FastAPI WebSockets, deployed with Docker containers.

### Developer Activity Data Platform (GitHub)

Python, Airflow, dbt, BigQuery, GCS, FastAPI, React

- Engineered an end-to-end data platform ingesting GitHub events via Airflow DAGs into BigQuery incrementally.
- Designed a multi-layer dbt pipeline (staging, facts/dims, marts) with partitioned tables and data quality tests.
- Exposed analytics via a FastAPI + React dashboard surfacing trending repos and language growth patterns.

### Financial Market Data Pipeline (GitHub)

Python, Airflow, dbt, Redshift, Terraform

- Engineered a batch financial data pipeline on AWS ingesting equity data through a medallion architecture.
- Built dbt models (staging, intermediate, marts) detecting pre-earnings volume anomalies across equities.
- Orchestrated five Airflow DAGs covering ingestion, validation, and transformation with Great Expectations.

## TECHNICAL SKILLS

**Languages:** Python, SQL, C#, Go, Bash, JavaScript, TypeScript

**Data Engineering:** Spark, Kafka, Airflow, dbt, BigQuery, Redshift, Pandas, NumPy, ETL/ELT, REST APIs

**Databases:** PostgreSQL, Redshift, BigQuery, MongoDB, SQLite, Oracle PL/SQL

**Cloud & DevOps:** Azure, AWS S3, GCS, Docker, Terraform, Git, CI/CD, Bicep, Linux

**AI & ML:** LangChain, LangGraph, Azure OpenAI, RAG Pipelines, Isolation Forest, k-means